

M.AI.ESTRO VR: system audit

Finalized 4 October 2026 · Graham Pellegrini · University of Malta

System Demo

Full recording stitch

[Open full recording](#)

Clock orbit: capture and replay

RGB / SAM2 on the left · Unity ghost replay on the right · Playback at 3× speed. [Open orbit comparison](#)

Clock task: capture and replay

RGB / SAM2 on the left · Unity ghost replay on the right. Task replay uses the latest orbit mesh under the temporary demo policy. [Open task comparison](#)

Pipeline Diagram

The pipeline PNG shows the flow from headset capture to stored object evidence, with separate task and orbit branches. Open the full-resolution image to inspect individual labels; the PDF gives the complete diagram its own landscape A3 page.

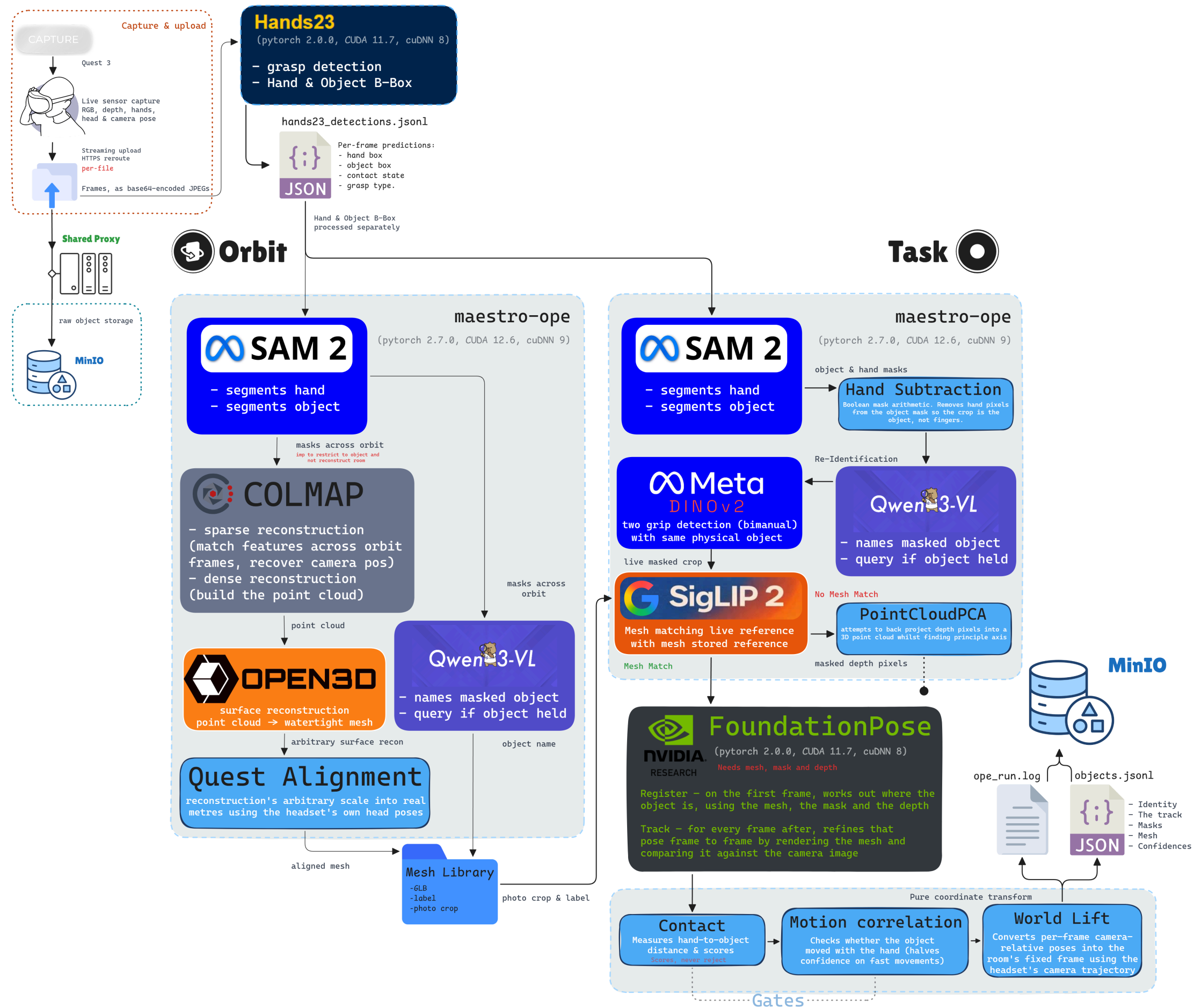


Figure 1. Current Maestros VR pipeline used for the audit discussion.

Pipeline Modules

Each node has a narrow responsibility. The audit question is whether the handoff between nodes is reliable enough for replay.

1

Quest capture

Records RGB, depth, headset pose and hand pose for task and orbit sessions.

2

Ingestion and MinIO

Accepts uploads, stores raw inputs and exposes session artifacts for worker processing.

3

Hands23 hand detection

Provides hand/object contact evidence and candidate boxes before object processing begins.

4

SAM2 segmentation

Turns candidate evidence into masks for orbit reconstruction and task-frame object tracking.

5a

PointCloudPCA

Provides model-free task pose estimates when a trusted mesh is not available.

5b

FoundationPose with mesh

Runs when a suitable mesh, mask and depth are available; this is the stronger route for mesh-backed replay.

6

Qwen labels

Supplies an object label from the crop; useful, but still dependent on mask quality.

7

DINOv2 and SigLIP

Support visual identity checks. The current audit keeps DINOv2 and SigLIP rather than adopting DINOv3.

8

COLMAP mesh reconstruction

Reconstructs object meshes from orbit capture frames for canonical mesh-backed replay.

9

Headset replay

Consumes stored objects, tracks, meshes and masks so captured objects can be inspected in VR.

System Status

All status items are shown together. The coloured badge carries the state; open a card for the full note.

Orbit capture route end-to-end



Hands23 → SAM2 → FoundationPose with mesh, full replay in headset.

— Modules 3 → 4 → 5b → 9 → status

Confirmed

FoundationPose (pose estimation with mesh)



Previously non-functional due to CUDA incompatibility and missing Docker configuration. Now properly containerised and running.

— Module 5b → status

Confirmed

SAM2 segmentation (replacing SAM2+SAM3)



Object segmentation now uses SAM2 exclusively; the previous SAM2/SAM3 dual setup has been simplified.

— Module 4 → status

Confirmed

Hands23 hand detection



Detects hand grip events and seeds object tracking; replaced a previous subtraction-based approach.

— Module 3 → status

Confirmed

COLMAP mesh reconstruction (orbit route)



Reconstructs a 3D mesh from orbit capture frames using photogrammetric reconstruction.

— Module 8 → status

Confirmed

Episode naming endpoint



Server generates structured display names (e.g. clock_00_261003) served by UUID; Unity displays verbatim.

— Module 2 → status

Confirmed

Mesh in headset replay



Object mesh renders in replay instead of wireframe for confirmed demo sessions.

— Module 9 → status

Fixed for demo

PointCloudPCA (task route pose fallback)



With mesh matching disabled, task sessions fall back to a per-frame point cloud centroid estimate with no temporal memory.

— Module 5a → status

Partially fixed

Mesh matching (automatic object identification)



Automatically matches a live object crop to a library mesh using SigLIP image embeddings.

— Module 7 → status

Disabled

Latest orbit mesh for task replay



Task replay uses the most recently registered orbit mesh directly, without automatic identity matching.

— Modules 7 → 9 → status

Fixed for demo

Replay orientation and interaction



User reports inconsistent ghost-hand facing, incorrect clock orientation and recording-stamp interaction problems.

— Module 9 → status

Not confirmed

DINOv3 encoder consolidation



Evaluated replacing SigLIP + DINOv2 with a single DINOv3 encoder to simplify the pipeline.

— Module 7 → status

Evaluated, not adopted

WiFi offline state (Unity client)



The Unity client shows connected and unstable WiFi states but has no fully-offline state.

— Module 9 → status

Not confirmed

Maestro server endpoint (shared proxy)



Maestro currently uses the shared Faye IT endpoint rather than its own dedicated server endpoint.

— Module 2 → status

Not confirmed

Technical Detail

[Full technical audit and measurements](#)

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